ELEC3104: Digital Signal Processing Lab

Topic 6- Multirate signal processing

MATLAB Examples

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Topic 6 – Decimation

Lab Example 1: **Sampling Rate Conversion of a Speech Signal**

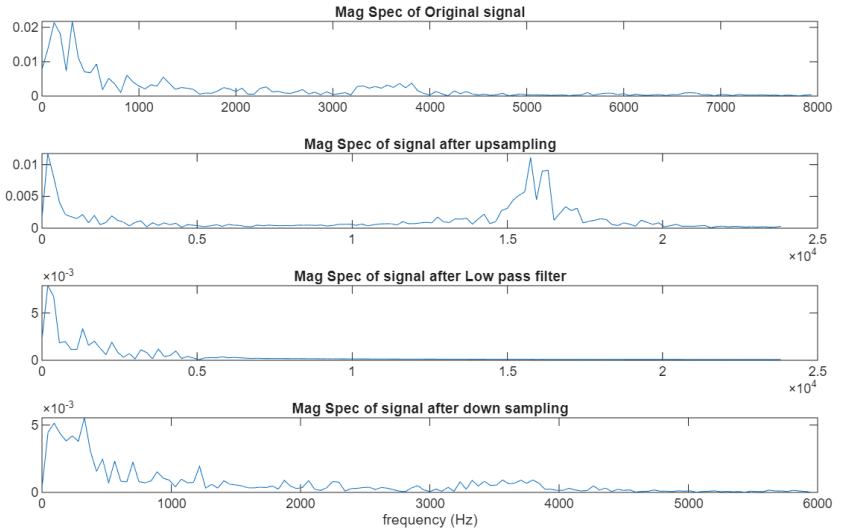
A speech signal sampled at 16 kHz is stored in speech.wav. You are required to convert the sampling rate from 16 kHz to 12 kHz using multirate signal processing.

1. Determine the interpolation factor L and decimation factor M required to convert the sampling frequency from 16 kHz to 12 kHz.
2. Draw a block diagram showing the complete sampling-rate conversion system, including:
   - interpolation
   - low-pass filtering
   - decimation.
3. Determine the sampling frequency at each stage of the system.
4. Determine the required cut-off frequency of the low-pass filter. Design an FIR low-pass filter in MATLAB and plot its magnitude response with the frequency axis in Hz.
5. Implement the complete sampling-rate conversion system in MATLAB.
6. Plot the magnitude spectra, with the frequency axis in Hz, of:
   - the original 16 kHz speech signal
   - the signal immediately after upsampling
   - the signal after low-pass filtering
   - the final 12 kHz speech signal.
7. Listen to the original and resampled speech signals. Comment briefly on any audible difference.
8. Explain the purpose of the low-pass filter. What would happen if the filtering operation were omitted?

Topic 6 – Decimation and Interpolation

```
% multirate resample speech from 16kHz to 12kHz
[x,fs]=audioread('speech.wav');
% sampling rate of x is at 16kHz;
% To convert to 12kHz, upsample by 3 (L), followed by lowpass filtering and
then downsample by 4 (M)
xUp=zeros(1,3*length(x));
xUp(1:3:end)=x; % upsampling by 3
fs_up = fs * L; % sampling frequency after interpolation
fs_out = fs_up / M; % sampling frequency after decimation
fc = 6000; % cut-off frequency
% lowpass filter design
Wn = fc/(fs_up/2);
B = fir1(40,Wn);
[H,w]=freqz(B,1);
plot((fs*3)*w/pi,abs(H));
xlabel('frequency (Hz)'); ylabel('|H(w)|')
grid on
xLP=filter(B,1,xUp); % lowpass filtering
xDown=xLP(1:4:end); % downsampling by 4
%plot magnitude spectrum and waveform
NFFT=256;
freqHz=(0:NFFT-1)/NFFT;
freqHz=freqHz(1:NFFT/2);
X=abs(fft(x,NFFT)); % magnitude of original speech
Xup=abs(fft(xUp,NFFT)); % magnitude of upsampled speech
XLP=abs(fft(xLP,NFFT)); % magnitude of LP filtered speech
XDown=abs(fft(xDown,NFFT)); % magnitude of downsampled speech
```

```
X=X(1:NFFT/2);
Xup=Xup(1:NFFT/2);
XDown=XDown(1:NFFT/2);
XLP=XLP(1:NFFT/2);
figure
subplot 411
plot(freqHz*fs,X); title('Mag Spec of Original signal')
subplot 412
plot(freqHz*3*fs,Xup); title('Mag Spec of signal after upsampling')
subplot 413
plot(freqHz*3*fs,XLP); title('Mag Spec of signal after Low pass filter')
subplot 414
plot(freqHz*3*fs/4,XDown); title('Mag Spec of signal after down sampling')
sound(xDown,fs*3/4)
% sound(x,fs)
xlabel('frequency (Hz)')
figure
subplot (211); plot(x); title('Waveform of original signal fs=16kHz')
subplot (212); plot(xDown); title('Waveform of resampled signal fs=12kHz')
```



Topic 6 – Decimation

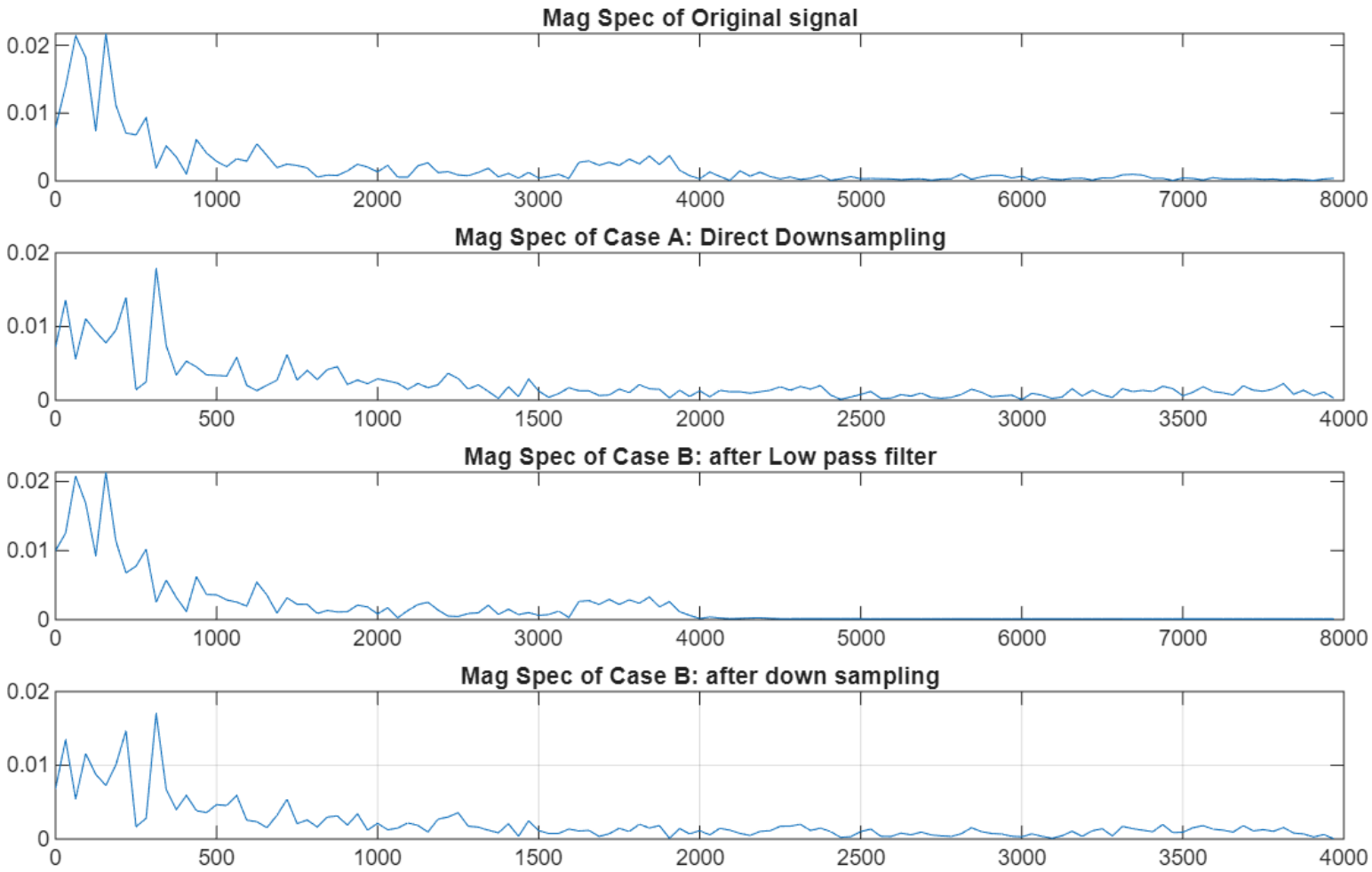
Lab Example 2: **Decimation of Speech: What Happens Without an Anti-Aliasing Filter?**

A speech signal is sampled at **16 kHz**. You wish to reduce the sampling rate to **8 kHz**.

1. Determine the required decimation factor M.
2. What is the highest frequency that can be represented without aliasing after decimation?
3. Using MATLAB, obtain the magnitude spectrum of the original speech signal.
4. Produce two 8 kHz signals:
   - **Case A:** Downsample the original speech directly by a factor of 2.
   - **Case B:** First low-pass filter the speech signal and then downsample it by a factor of 2.
5. Plot and compare the magnitude spectra of the signals obtained in Cases A and B.
6. Listen to both signals and comment on any audible differences.
7. Explain why the low-pass filter is called an anti-aliasing filter.

```
[x, fs] = audioread('speech.wav'); % Read speech signal
fprintf('Original sampling frequency = %d Hz\n', fs);
M = 2; % Decimation factor
yA = x(1:M:end); % Case A: Direct downsampling
% Case B: Low-pass filter then downsample
fc = 4000; % Cutoff frequency in Hz
Nf = 40; % FIR filter order
Wn = fc/(fs/2);
b = fir1(Nf, Wn);
x_filt = filter(b, 1, x); % low-pass filtering the original 16 kHz speech
yB = x_filt(1:M:end); % Downsample by 2
% plot magnitude spectrum and waveform
NFFT=256;
freqHz=(0:NFFT-1)/NFFT;
freqHz=freqHz(1:NFFT/2);
X=abs(fft(x,NFFT));
YA=abs(fft(yA,NFFT));
X_filt=abs(fft(x_filt,NFFT));
YB=abs(fft(yB,NFFT));
X=X(1:NFFT/2);
YA=YA(1:NFFT/2);
X_filt=X_filt(1:NFFT/2);
YB=YB(1:NFFT/2);
figure
subplot (411); plot(freqHz*fs,X); title('Mag Spec of Original signal')
subplot (412); plot(freqHz*fs/M,YA); title('Mag Spec of Case A: Direct down
sampling')
subplot (413); plot(freqHz*fs,X_filt); title('Mag Spec of Case B: after Low pass filter')
subplot (414); plot(freqHz*fs/M,YB); title('Mag Spec of Case B: after down
sampling');
grid on;
fs_new = fs/M;
% Listen to Case A
sound(yA, fs_new);
pause(length(yA)/fs_new + 1);
% Listen to Case B
sound(yB, fs_new);
```

Topic 6 – Decimation



Topic 6 –Interpolation

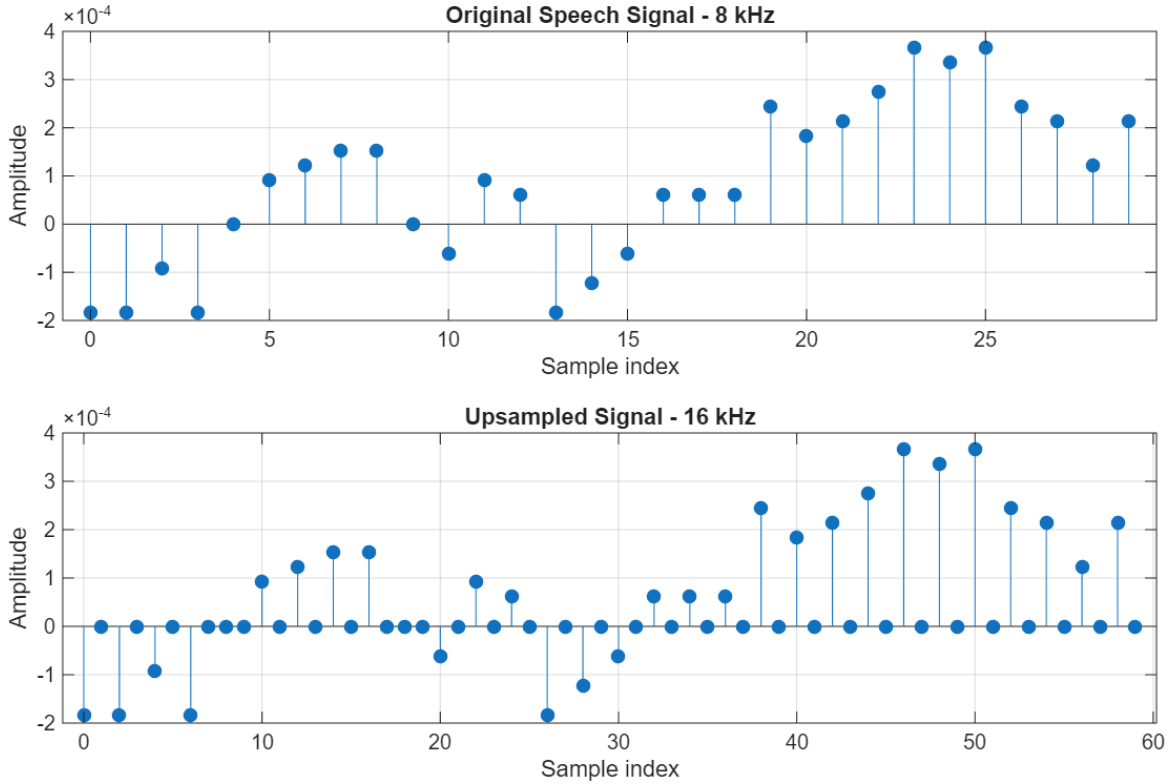
Lab Example 3: **Interpolation of a Speech Signal**

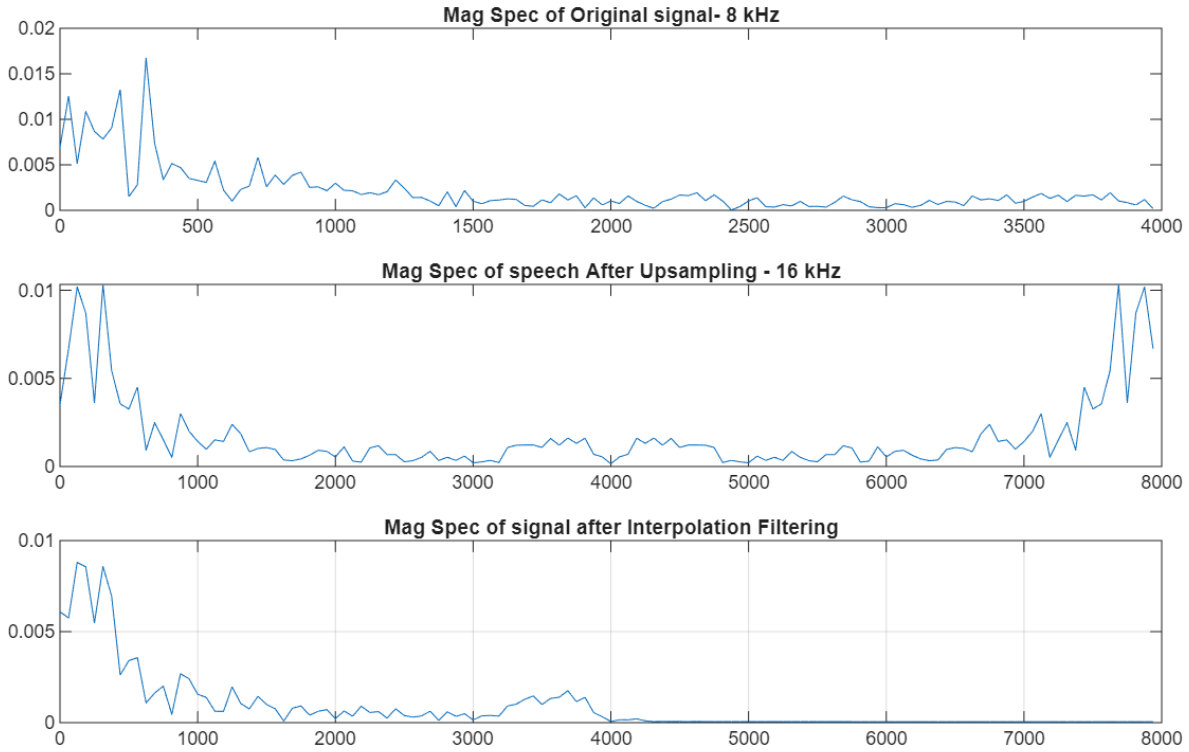
A speech signal is sampled at **8 kHz**. You are required to increase its sampling rate to **16 kHz**.

1. Determine the interpolation factor L.
2. Upsample the signal by inserting the required number of zeros between consecutive samples.
3. Plot a short section of the original and upsampled signals in the time domain.
4. Plot the magnitude spectrum of the upsampled signal. What new spectral components do you observe?
5. Design an appropriate FIR low-pass interpolation filter and filter the upsampled signal.
6. Plot the magnitude spectrum after filtering and compare it with the spectrum before filtering.
7. Listen to the interpolated signal.
8. Explain the purpose of the interpolation filter.

```
[x, fs] = audioread('speech_8K.wav'); % Read speech signal
L = 2; % Interpolation factor
fs_up = L*fs;
x_up = zeros(1, L*length(x)); % UPSAMPLING BY ZERO INSERTION
x_up(1:L:end) = x; % upsampling by L
% Time-domain comparison
Ns = 30; % Number of original samples to display
n1 = 0:Ns-1;
n2 = 0:L*Ns-1;
figure;
subplot(2,1,1); stem(n1, x(1:Ns), 'filled');
xlabel('Sample index'); ylabel('Amplitude'); title('Original Speech Signal - 8 kHz'); grid on;
subplot(2,1,2); stem(n2, x_up(1:L*Ns), 'filled');
xlabel('Sample index'); ylabel('Amplitude'); title('Upsampled Signal - 16 kHz'); grid on;
% FIR INTERPOLATION FILTER
fc = 4000; % cutoff frequency
Nf = 80; % FIR filter order
Wn = fc/(fs_up/2);
b = fir1(Nf, Wn);
x_filt_up = filter(b, 1, x_up); % Apply the FIR interpolation filter to the upsampled signal
% plot magnitude spectrum and waveform
NFFT=256;
freqHz=(0:NFFT-1)/NFFT;
freqHz=freqHz(1:NFFT/2);
X=abs(fft(x,NFFT));
X_up=abs(fft(x_up,NFFT));
X_filt_up=abs(fft(x_filt_up,NFFT));
X=X(1:NFFT/2);
X_up=X_up(1:NFFT/2);
X_filt_up=X_filt_up(1:NFFT/2);
figure
subplot (311); plot(freqHz*fs,X); title('Mag Spec of Original signal- 8 kHz');
subplot (312); plot(freqHz*fs_up,X_up); title('Mag Spec of speech After Upsampling - 16 kHz');
subplot (313); plot(freqHz*fs_up,X_filt_up); title('Mag Spec of signal after Interpolation Filtering'); grid on;
% Listen to original signal
sound(x, fs);
pause(length(x)/fs + 1);
% Listen to interpolated signal
sound(x_filt_up, fs_up);
```

Topic 6 –Interpolation





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Lab Example 4: **Effect of Different Decimation Factors**

A speech signal is sampled at **16 kHz**. Investigate what happens when its sampling rate is progressively reduced.

1. Using appropriate anti-aliasing filters, generate versions of the speech signal sampled at:
    - 8 kHz
    - 4 kHz
    - 2 kHz
2. For each case:
    - determine the decimation factor,
    - determine the required anti-aliasing filter cut-off frequency,
    - plot the magnitude spectrum, and
    - listen to the resulting speech.
3. Compare the quality and intelligibility of the speech at the four sampling rates: 16, 8, 4 and 2 kHz.
4. Explain why speech quality changes as the sampling frequency is reduced.

Topic 6 –Interpolation

```
[x, fs] = audioread('speech.wav');
N0 = length(x);
X0 = fft(x);
X0mag = abs(X0(1:floor(N0/2)+1));
f0 = (0:floor(N0/2))*fs/N0;
% CASE 1: 16 kHz -> 8 kHz
M1 = 2;
fs1 = fs/M1;
fc1 = 4000; % cutt-off frequency 4 kHz
Nf = 80; % FIR filter order
b1 = fir1(Nf, fc1/(fs/2)); % FIR filter
x1_filt = filter(b1, 1, x); % filtering
x8 = x1_filt(1:M1:end);
N1 = length(x8);
X1 = fft(x8);
X1mag = abs(X1(1:floor(N1/2)+1));
f1 = (0:floor(N1/2))*fs1/N1;
% CASE 2: 16 kHz -> 4 kHz
M2 = 4;
fs2 = fs/M2;
fc2 = 2000; % cutt-off frequency 2 kHz
b2 = fir1(Nf, fc2/(fs/2));
x2_filt = filter(b2, 1, x);
x4 = x2_filt(1:M2:end);
N2 = length(x4);
X2 = fft(x4);
X2mag = abs(X2(1:floor(N2/2)+1));
f2 = (0:floor(N2/2))*fs2/N2;
```

```
% CASE 3: 16 kHz -> 2 kHz
M3 = 8;
fs3 = fs/M3;
fc3 = 1000; % cutt-off frequency 1 kHz
b3 = fir1(Nf, fc3/(fs/2));
x3_filt = filter(b3, 1, x);
x2 = x3_filt(1:M3:end);
N3 = length(x2);
X3 = fft(x2);
X3mag = abs(X3(1:floor(N3/2)+1));
f3 = (0:floor(N3/2))*fs3/N3;

% PLOT MAGNITUDE SPECTRA
figure;
subplot(4,1,1);
plot(f0, 20*log10(X0mag));
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Original Speech - 16 kHz');
grid on;
xlim([0 fs/2]);
subplot(4,1,2);
plot(f1, 20*log10(X1mag));
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Speech - 8 kHz');
grid on;
xlim([0 fs1/2]);
```

```
subplot(4,1,3);
plot(f2, 20*log10(X2mag));
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Speech - 4 kHz');
grid on;
xlim([0 fs2/2]);
subplot(4,1,4);
plot(f3, 20*log10(X3mag));
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Speech - 2 kHz');
grid on;
xlim([0 fs3/2]);
% Listen to original
sound(x, fs);
pause(length(x)/fs + 1);
% Listen to 8 kHz speech
sound(x8, fs1);
pause(length(x8)/fs1 + 1);
% Listen to 4 kHz speech
sound(x4, fs2);
pause(length(x4)/fs2 + 1);
% Listen to 2 kHz speech
sound(x2, fs3);
```

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